

Multiple Choice

1. **Answer:** A

Options: A) 11.3; B) 7.0; C) 13.7; D) 6.5; E) 12.5

Note: Correct option: A - 11.3

2. **Answer:** A

Options: A) volume; B) timbre; C) speed; D) duration; E) wavelength

Note: Correct option: A - volume

3. **Answer:** E

Options: A) 5.0×10^3 cal; B) 15.3×10^3 cal; C) 10.5×10^3 cal; D) 12.7×10^3 cal; E) 7.9×10^3 cal

Note: Correct option: E - 7.9×10^3 cal

4. **Answer:** A

Options: A) $6.7 \times 10^{-17}kg$; B) $2.2 \times 10^{-22}kg$; C) $3.0 \times 10^{-32}kg$; D) $5.5 \times 10^{-31}kg$; E) 1.76×10^{11} kg

Note: Correct option: A - $6.7 \times 10^{-17}kg$

4. **Answer:** A

Options: A) remains the same; B) increases by 4; C) reduces by 3; D) increases by 3; E) increases by 1

Note: Correct option: A - remains the same

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '5.0 N'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 8.660×10^6 m. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $\frac{1}{3}u_1$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key